MOBILE APPLICATION FOR BUYING AND SELLING BACKYARD SWINE

System Development Documentation

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INTRODUCTION

Project Overview

This documentation covers the development of the Mobile Application for Buying and Selling Backyard Swine, a capstone project designed to help backyard swine farmers in Quezon, Bukidnon manage their livestock, connect with buyers, and access important farming information through their mobile phones.

The application was built using React Native for cross-platform mobile development, JavaScript as the main programming language, Supabase for backend services and database management, and Antigravity as the development environment. The system includes features like user management, geospatial mapping, marketplace functionality, feeding information, and health guidelines based on Department of Agriculture standards.

Team Roles and Responsibilities

Team Member	Primary Roles	Responsibilities
JOESA MAE FARIOLAN	● Project Manager ● Lead Programmer ● Technical Writer	• Overall project coordination and timeline management • Core programming using React Native and JavaScript • Writing technical documentation and user manuals • Sprint planning, task delegation, progress tracking • Git repository management and version control • Bug fixing, code optimization, performance tuning • Deployment preparation and release management
SHERLEN ME PEROCHO	● UI/UX Designer ● Frontend Developer ● Quality Assurance Tester	● User interface design, wireframing, and prototyping ● Frontend component development in React Native ● Color scheme, typography, icon design, visual branding

		● Screen layout, navigation flow, responsive design ● User experience testing and feedback collection ● Usability testing coordination and SUS administration ● Quality assurance, bug reporting, regression testing ● Design consistency checks across all app screens
ANNA JANE ESPRA	● Database Administrator ● Research Coordinator ● Backend Developer	● Database design, schema creation, and ERD development ● Supabase database configuration, management, optimization ● Data security setup, access control, RLS policies ● Writing database queries, functions, real-time subscriptions ● User authentication configuration and security implementation ● System requirements analysis and data flow diagrams ● Research coordination, literature review, data gathering

		● Data backup planning, performance monitoring analytics

Project Scope

The project focuses on developing a mobile application specifically for backyard pig farmers in Quezon, Bukidnon. The application covers the following main modules:

- User Management Module – Registration, login, and role-based access for farmers, buyers, and admin users.
- Database Management Module – Secure storage of user profiles, livestock records, marketplace listings, and feeding information using Supabase Database.
- Geospatial Mapping Module – Visualization of farm locations using ESRI and OpenStreetMap for better accessibility and logistics.
- Marketplace Module – Platform for farmers to list livestock for sale and for buyers to browse and contact sellers directly.
- Feeding Information Module – Guidelines based on Department of Agriculture standards for proper swine feeding practices.
- Health Guidelines Module – Information on common swine diseases, symptoms, prevention, and recommended actions.

The study does not cover commercial livestock operations, real-time disease detection, or predictive analytics. The application's effectiveness depends on user adoption, digital literacy, and internet connectivity in rural areas.

SYSTEM ARCHITECTURE

Technology Stack

The development team chose the following technology stack based on the project's requirements for cross-platform compatibility, real-time data handling, and ease of use for farmers with varying levels of technical experience.

Technology	Purpose	Responsible	Description
React Native	Mobile Framework	Joesa Mae	Cross-platform framework by Meta for building native-like mobile apps using JavaScript. Single

			codebase for Android and iOS.
JavaScript	Programming Language	Joesa Mae	Main language used with React Native. Flexible, widely-supported, with large developer community.
Supabase	Backend Service	Anna Jane	Open-source alternative to Firebase with PostgreSQL database, real-time subscriptions, auth, and storage.
Supabase Auth	Authentication	Anna Jane	Secure login with email/password, phone OTP, and social login providers.
Supabase DB	Database	Anna Jane	PostgreSQL-based database with real-time capabilities for all application data.
OpenStreetMap	Mapping	Sherlen	Free, open-source mapping for road networks, navigation, and directions.
ESRI	Satellite Maps	Sherlen	Commercial GIS with high-quality satellite imagery for visualizing farm locations.
Antigravity	Code Editor	Joesa Mae	AI-powered editor by Google based on VS Code with AI agents for coding assistance.

System Components

The system follows a client-server architecture where the mobile application (client) communicates with the Supabase backend (server) to perform operations.

Client-Side (Mobile Application):

- Built with React Native and JavaScript
- Handles user interface rendering and user interactions
- Communicates with Supabase via REST API and real-time subscriptions
- Stores minimal local data for offline capability
- Uses GPS for location tracking and mapping features

Server-Side (Supabase Backend):

- PostgreSQL database for structured data storage
- Authentication service for secure user login and registration
- Storage bucket for livestock images and user profile photos
- Real-time subscriptions for instant data updates
- Row Level Security (RLS) for data protection
- Edge Functions for server-side logic when needed

DATABASE DESIGN

This chapter discusses the database design for the Backyard Swine Farming Application. The database was designed and managed by Anna Jane Espra using Supabase Database (PostgreSQL).

Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) shows how the different data entities in the system relate to each other. The database is structured around the User entity, which connects to all other modules.

Key Entities:

- User – Stores account information (name, email, password, role: admin/buyer/seller)
- Livestock – Records details of each animal (type, breed, age, weight, status, owner)
- Marketplace – Lists livestock for sale with price, description, and status
- GeospatialMapping – Stores latitude and longitude coordinates for map display
- FeedingInformation – Contains feeding guidelines linked to livestock records
- Health – Tracks health records and disease information per livestock
- Logs – Records user actions for system monitoring and audit trails

Relationships:

- One User can have many Livestock records (1:N)
- One Livestock can have one Marketplace listing at a time (1:1)
- One Livestock can have many FeedingInformation entries (1:N)
- One User has one GeospatialMapping entry (1:1)
- One User can have many Logs entries (1:N)

Database Schema

The database schema was implemented in Supabase Database using PostgreSQL. Below are the main tables:

Table Name	Primary Key	Foreign Keys	Main Fields
User	UserID (INT)	None	Name, Email, Password, Role, CreatedAt
Livestock	LivestockID (INT)	UserID -> User	Type, Breed, Age, Weight, Location, Status
Marketplace	ListingID (INT)	LivestockID -> Livestock, SellerID -> User	Price, Description, Status, CreatedAt
GeospatialMapping	MappingID (INT)	UserID -> User	Latitude, Longitude, ViewType
FeedingInformation	InfoID (INT)	LivestockID -> Livestock	Guideline, Source (default: DOA)
Health	HealthID (INT)	LivestockID -> Livestock	Condition, Symptoms, Treatment, DateRecorded
Logs	LogID (INT)	UserID -> User	Action, Timestamp, Details

Data Dictionary

The data dictionary ensures consistent and accurate data management. It defines the structure, format, and purpose of each data element.

User Table:

Column	Data Type	Description	Constraints
UserID	INT	Unique ID for each user	Primary Key

Name	VARCHAR(255)	Full name of the user	Not Null
Email	VARCHAR(255)	Email address	Unique, Not Null
Password	VARCHAR(255)	Hashed password	Not Null
Role	ENUM	User type: Admin, Seller, Buyer	Not Null
CreatedAt	TIMESTAMP	Time account was created	Default: Current Time

Livestock Table:

Column	Data Type	Description	Constraints
LivestockID	INT	Unique ID for livestock	Primary Key
UserID	INT	Owner's user ID	Foreign Key -> User
Type	VARCHAR(255)	Type of livestock (e.g., pig)	Not Null
Breed	VARCHAR(255)	Breed of the livestock	Nullable
Age	INT	Age in years	Nullable
Weight	DECIMAL(10,2)	Weight in kilograms	Nullable
Location	VARCHAR(255)	Physical location	Nullable
Status	ENUM	Available, Sold, Unavailable	Default: Available

Marketplace Table:

Column	Data Type	Description	Constraints
ListingID	INT	Unique ID for each listing	Primary Key
LivestockID	INT	Livestock being sold	Foreign Key -> Livestock
SellerID	INT	Seller's user ID	Foreign Key -> User
Price	DECIMAL(10,2)	Price of livestock	Not Null
Description	TEXT	Description of the animal	Nullable
Status	ENUM	Available, Sold	Default: Available
CreatedAt	TIMESTAMP	Listing creation time	Default: Current Time

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USER INTERFACE DESIGN

This chapter covers the user interface design of the application, handled by Sherlen Me Perocho. The design focuses on simplicity and ease of use for farmers who may have limited experience with mobile apps.

Design Principles

- Simplicity – Clean layouts with minimal clutter to avoid confusion.
- Consistency – Same colors, fonts, and button styles throughout the app.
- Readability – Large fonts and high contrast for easy reading under sunlight.
- Accessibility – Touch targets are large enough for users with less precise motor control.
- Visual Hierarchy – Important information is prominently displayed.
- Feedback – Clear messages when actions are completed or errors occur.
- Localization – Content in simple English and local terms familiar to farmers.

Screen Descriptions

Login and Sign Up Screen: Allows users to register or log in using email and password. Admin accounts are pre-registered in the database. Simple form with clear labels and validation messages.

Dashboard or Home Screen: Main navigation hub showing quick access to livestock records, marketplace, map, feeding info, and health guidelines. Designed with large icon buttons for easy tapping.

Marketplace Screen: Displays livestock listings with images, prices, and seller info. Buyers can browse, search, and contact sellers directly. Filter options for location and price range.

Geospatial Map Screen: Interactive map showing farm locations using OpenStreetMap and ESRI satellite view. Different views for admin (all farms), sellers (own location), and buyers (nearby farms).

Feeding Information Screen: Department of Agriculture feeding guidelines presented in simple, step-by-step format. Includes visual aids and feeding schedules for different pig categories.

Health Guidelines Screen: Information on common swine diseases like African Swine Fever (ASF), symptoms, prevention methods, and recommended actions. Organized by disease type for easy reference.

Admin Panel: Administrator interface for managing user accounts, approving registrations, updating feeding and health content, and viewing system reports and geospatial data.

DEVELOPMENT PROCESS

This chapter describes how the team developed the application using Agile methodology, led by Joesa Mae Fariolan as Project Manager.

Agile Methodology

The team used Agile Development Methodology, which emphasizes iterative progress, flexibility, and continuous feedback. The project was divided into short development cycles called sprints, typically lasting two to four weeks.

Why Agile was chosen:

- Allows frequent testing and feedback from potential users (farmers).
- Easy to adapt to changes in requirements during development.
- Team members can focus on specific features one at a time.
- Early detection of problems through regular sprint reviews.
- Stakeholders (like the Municipality Agriculture Office) can see progress regularly.

Sprint Planning

The development was divided into 6 sprints, each focusing on a specific module:

Sprint 1: User Management Module (Weeks 1-4)

Team: Joesa Mae Fariolan (Lead), Anna Jane Espra (DB Support)
User registration and login forms
Supabase Authentication setup
Role-based access control (Admin, Seller, Buyer)
User profile management
Testing login flows and security

Sprint 2: Database Setup (Weeks 5-7)

Team: Anna Jane Espra (Lead), Joesa Mae Fariolan (Integration)
Database schema design in Supabase
Table creation and relationships
Row Level Security policies
Real-time subscription setup
Data validation rules
API endpoint configuration

Sprint 3: Geospatial Mapping (Weeks 8-10)
Team: Sherlen Me Perocho (Lead), Joesa Mae Fariolan (Integration)
OpenStreetMap integration
ESRI satellite view setup
GPS location capture
Map markers for farm locations
Different map views per user role
Navigation and directions feature

Sprint 4: Marketplace Module (Weeks 11-14)
Team: Joesa Mae Fariolan (Lead), Sherlen Me Perocho (UI)
Livestock listing creation
Image upload to Supabase Storage
Browse and search functionality
Contact seller feature
Price filtering and sorting
Transaction status tracking

Sprint 5: Feeding Information (Weeks 15-16)
Team: Anna Jane Espra (Content), Sherlen Me Perocho (Design)
Content compilation from DA guidelines
Database seeding with feeding data
User-friendly information display
Categorization by pig type and age
Visual aids and diagrams

Sprint 6: Health Guidelines and Testing (Weeks 17-20)
Team: All Team Members
Health content from DA sources
Disease information database
Symptom checker feature
Prevention tips display
System-wide testing
Bug fixing and optimization
Usability testing preparation

5.3 Communication and Collaboration

The team maintained regular communication through:

- Daily Stand-up Meetings – Quick updates on what was done, what will be done, and any blockers.
- Sprint Planning Meetings – At the start of each sprint to define tasks and assign responsibilities.
- Sprint Review Meetings – Demonstrating completed features to the team and advisor.
- Sprint Retrospectives – Discussing what went well and what needs improvement.
- Discord or Group Chat – For quick questions and file sharing.
- GitHub – For version control, code reviews, and issue tracking.

TESTING AND QUALITY ASSURANCE

This chapter covers the testing strategies and quality assurance processes led by Sherlen Me Perocho, with support from the whole team.

Testing Strategy

The application underwent multiple levels of testing to ensure reliability, security, and usability.

Types of Testing Performed:

Test Type	Purpose	Team Member	Tools Used
Unit Testing	Test individual components and functions	Joesa Mae Fariolan	Jest, React Native Testing Library
Integration Testing	Test how modules work together	Joesa Mae Fariolan	Jest, Supabase Test Environment
System Testing	Test the complete application flow	All Members	Manual testing, Test cases
Usability Testing	Evaluate ease of use for farmers	Joesa Mae Fariolan	System Usability Scale (SUS)
Security Testing	Check for vulnerabilities	Anna Jane Espra	Supabase RLS validation, Auth testing
Performance Testing	Check app speed and responsiveness	Anna Jane Espra	React Native Profiler, Supabase logs
User Acceptance Testing	Validate with actual farmers	All Members	Field testing in Quezon barangays

Usability Testing Results

Usability testing was conducted using the System Usability Scale (SUS) questionnaire with 170 backyard swine farmers from different barangays in Quezon, Bukidnon.

Key Results:

- Overall SUS Score: 63.12 (Marginal to Acceptable usability)
- Strengths: Marketplace and livestock management modules were rated easy to use.
- Areas for Improvement: Learnability, interface consistency, and guidance for geospatial mapping.
- Demographics: Majority of respondents were middle-aged (45-55 years old) and female (70%).

Interpretation:

The SUS score of 63.12 indicates that users generally found the application usable, but there is room for improvement. The team recommends adding onboarding tutorials, simplifying navigation, and providing better guidance for first-time users.

DEPLOYMENT AND MAINTENANCE

Deployment Plan

The deployment of the application follows a structured approach in close coordination with the Department of Agriculture (DA) to ensure a smooth and effective rollout to the target users.

- Phase 1: Internal Testing — Complete testing within the development team and with the project advisor to ensure core functionalities are stable and ready for stakeholder handover.
- Phase 2: Stakeholder Presentation & Turnover — Present the finalized application to the DA, conduct a walkthrough of its features, and officially turn over the system, including all documentation, source code, and deployment assets.
- Phase 3: Pilot Testing — Collaborate with the DA to deploy the application in selected barangays (Salawagan, San Jose, Kiburiao) for initial field testing and feedback collection from early-adopter farmers.
- Phase 4: Training of Trainers (ToT) — Conduct capacity-building sessions for designated DA agricultural extension workers and local technicians, equipping them to train farmers and provide first-level support.
- Phase 5: Farmer Training & Orientation — In partnership with the DA, organize user orientation and training workshops for farmers across all target barangays, facilitated by the trained extension workers.
- Phase 6: Full Rollout — The DA formally distributes and deploys the application to all target barangays in Quezon, Bukidnon, leveraging their existing networks and farmer associations.

- Phase 7: Monitoring, Support & Feedback Loop — Provide ongoing technical support to the DA team, monitor app usage, and establish a structured feedback mechanism to inform future improvements.

Maintenance Strategy

Regular maintenance is essential to keep the application running smoothly and to address feedback gathered through the DA and end users.

- Bug Fixes — Address reported issues within one week of identification, with critical bugs escalated immediately.
- Feature Updates — Add new features based on requests from the DA and farmers, aligned with evolving agricultural needs.
- Security Updates — Regularly update Supabase security policies, access controls, and system dependencies.
- Data Backups — Automated daily backups of the Supabase database to ensure data integrity and disaster recovery.
- User Support — Maintain a dedicated help desk or contact point to assist the DA support team and, by extension, the farmers.
- Performance Monitoring — Continuously track app usage, response times, and system health to identify and resolve bottlenecks proactively.

Glossary

React Native: A cross-platform mobile development framework by Meta that uses JavaScript to build native-like apps for Android and iOS.

JavaScript: A popular programming language used for web and mobile application development.

Supabase: An open-source backend-as-a-service platform that provides database, authentication, and storage services.

Supabase Database: A PostgreSQL-based database service provided by Supabase with real-time capabilities.

Supabase Authentication: A secure login service by Supabase supporting email, phone, and social provider authentication.

Antigravity: An AI-powered code editor by Google based on Visual Studio Code with automated coding assistance.

Agile Methodology: A project management approach that uses iterative development cycles called sprints.

SUS (System Usability Scale): A standardized questionnaire used to measure the usability of a system.

ERD (Entity-Relationship Diagram): A visual representation of the database structure showing entities and their relationships.

API (Application Programming Interface): A set of rules that allows different software applications to communicate with each other.

RLS (Row Level Security): A database security feature that controls access to rows in a table based on user roles.

GPS (Global Positioning System): A satellite-based navigation system used for location tracking.

GIS (Geographic Information System): A system for capturing, storing, checking, and displaying data related to positions on Earth's surface.

Backend: The server-side part of an application that handles data storage, processing, and business logic.

Frontend: The client-side part of an application that users interact with directly.